

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

A bag contains 8 balls, whose colours are either white or black. 4 balls are drawn at random without replacement and it was found that 2 balls are white and other 2 balls are black. The probability that the bag contains equal number of white and black balls is:

(1) $\frac{2}{5}$

(2) $\frac{2}{7}$

(3) $\frac{1}{7}$

(4) $\frac{1}{5}$

Q2 - 2024 (01 Feb Shift 2)

Let Ajay will not appear in JEE exam with probability $p = \frac{2}{7}$, while both Ajay and Vijay will appear in the exam with probability $q = \frac{1}{5}$. Then the probability, that Ajay will appear in the exam and Vijay will not appear is :

(1) $\frac{9}{35}$

(2) $\frac{18}{35}$

(3) $\frac{24}{35}$

(4) $\frac{3}{35}$

Q3 - 2024 (27 Jan Shift 1)

A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required and let

$a = P(X = 3)$, $b = P(X \geq 3)$ and $c = P(X \geq 6 \mid X > 3)$. Then $\frac{b+c}{a}$ is equal to _____

Q4 - 2024 (27 Jan Shift 2)

An urn contains 6 white and 9 black balls. Two successive draws of 4 balls are made without replacement. The probability, that the first draw gives all white balls and the second draw gives all black balls, is :

(1) $\frac{5}{256}$

(2) $\frac{5}{715}$

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(3) $\frac{3}{715}$

(4) $\frac{3}{256}$

Q5 - 2024 (29 Jan Shift 1)

A fair die is thrown until 2 appears. Then the probability, that 2 appears in even number of throws, is

(1) $\frac{5}{6}$

(2) $\frac{1}{6}$

(3) $\frac{5}{11}$

(4) $\frac{6}{11}$

Q6 - 2024 (29 Jan Shift 2)

An integer is chosen at random from the integers 1, 2, 3, ..., 50. The probability that the chosen integer is a multiple of atleast one of 4, 6 and 7 is

(1) $\frac{8}{25}$

(2) $\frac{21}{50}$

(3) $\frac{9}{50}$

(4) $\frac{14}{25}$

Q7 - 2024 (30 Jan Shift 1)

Two integers x and y are chosen with replacement from the set $\{0, 1, 2, 3, \dots, 10\}$. Then the probability that

$|x - y| > 5$ is :

(1) $\frac{30}{121}$

(2) $\frac{62}{121}$

(3) $\frac{60}{121}$

(4) $\frac{31}{121}$

Q8 - 2024 (30 Jan Shift 2)

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Bag A contains 3 white, 7 red balls and bag B contains 3 white, 2 red balls. One bag is selected at random and a ball is drawn from it. The probability of drawing the ball from the bag A, if the ball drawn is white, is :

(1) $\frac{1}{4}$

(2) $\frac{1}{9}$

(3) $\frac{1}{3}$

(4) $\frac{3}{10}$

Q9 - 2024 (31 Jan Shift 1)

Two marbles are drawn in succession from a box containing 10 red, 30 white, 20 blue and 15 orange marbles, with replacement being made after each drawing. Then the probability, that first drawn marble is red and second drawn marble is white, is

(1) $\frac{2}{25}$

(2) $\frac{4}{25}$

(3) $\frac{2}{3}$

(4) $\frac{4}{75}$

Q10 - 2024 (31 Jan Shift 1)

Three rotten apples are accidentally mixed with fifteen good apples. Assuming the random variable x to be the number of rotten apples in a draw of two apples, the variance of x is

(1) $\frac{37}{153}$

(2) $\frac{57}{153}$

(3) $\frac{47}{153}$

(4) $\frac{40}{153}$

Q11 - 2024 (31 Jan Shift 2)

A coin is biased so that a head is twice as likely to occur as a tail. If the coin is tossed 3 times, then the probability of getting two tails and one head is-

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Answer Key

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Solutions

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Q1

$$\begin{aligned}
 P(4W4B/2W2B) &= \frac{P(4W4B) \times P(2W2B/4W4B)}{P(2W6B) \times P(2W2B/2W6B) + P(3W5B) \times P(2W2B/3W5B) + \dots + P(6W2B) \times P(2W2B/6W2B)} \\
 &= \frac{\frac{1}{5} \times \frac{{}^4C_2 \times {}^4C_2}{{}^8C_4}}{\frac{1}{5} \times \frac{{}^2C_2 \times {}^6C_2}{{}^8C_4} + \frac{1}{5} \times \frac{{}^3C_2 \times {}^5C_2}{{}^8C_4} + \dots + \frac{1}{5} \times \frac{{}^6C_2 \times {}^2C_2}{{}^8C_4}} \\
 &= \frac{2}{7}
 \end{aligned}$$

Q2



$$P(\bar{A}) = \frac{2}{7} = p$$

$$P(A \cap V) = \frac{1}{5} = q$$

$$P(A) = \frac{5}{7}$$

$$\text{Ans. } P(A \cap \bar{V}) = \frac{18}{35}$$

Q3

$$\begin{aligned}
 a = P(X = 3) &= \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \frac{25}{216} \\
 b = P(X \geq 3) &= \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} + \dots \\
 &= \frac{\frac{25}{216}}{1 - \frac{5}{6}} = \frac{25}{216} \times \frac{6}{1} = \frac{25}{36} \\
 P(X \geq 6) &= \left(\frac{5}{6}\right)^5 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^6 \cdot \frac{1}{6} + \dots
 \end{aligned}$$

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Solutions

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$$\begin{aligned}
 &= \frac{\left(\frac{5}{6}\right)^5 \cdot \frac{1}{6}}{1 - \frac{5}{6}} = \left(\frac{5}{6}\right)^5 \\
 &c = \frac{\left(\frac{5}{6}\right)^5}{\left(\frac{5}{6}\right)^3} = \frac{25}{36} \\
 &\frac{b+c}{a} = \frac{\left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^2}{\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}} = 12
 \end{aligned}$$

Q4

$$\frac{{}^6C_4}{{}^{15}C_4} \times \frac{{}^9C_4}{{}^{11}C_4} = \frac{3}{715}$$

Hence option (3) is correct.

Q5

Required probability =

$$\begin{aligned}
 &\frac{5}{6} \times \frac{1}{6} + \left(\frac{5}{6}\right)^3 \times \frac{1}{6} + \left(\frac{5}{6}\right)^5 \times \frac{1}{6} + \dots \\
 &= \frac{1}{6} \times \frac{\frac{5}{6}}{1 - \frac{25}{36}} = \frac{5}{11}
 \end{aligned}$$

Q6

Given set = {1, 2, 3, 50}

P(A) = Probability that number is multiple of 4

P(B) = Probability that number is multiple of 6

P(C) = Probability that number is multiple of 7

Now,

$$P(A) = \frac{12}{50}, P(B) = \frac{8}{50}, P(C) = \frac{7}{50}$$

again

$$P(A \cap B) = \frac{4}{50}, P(B \cap C) = \frac{1}{50}, P(A \cap C) = \frac{1}{50}$$

$$P(A \cap B \cap C) = 0$$

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Solutions

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Thus

$$P(A \cup B \cup C) = \frac{12}{50} + \frac{8}{50} + \frac{7}{50} - \frac{4}{50} - \frac{1}{50} - \frac{1}{50} + 0$$

$$= \frac{21}{50}$$

Q7

If $x = 0, y = 6, 7, 8, 9, 10$ If $x = 1, y = 7, 8, 9, 10$ If $x = 2, y = 8, 9, 10$ If $x = 3, y = 9, 10$ If $x = 4, y = 10$ If $x = 5, y =$ no possible value

$$\text{Total possible ways} = (5 + 4 + 3 + 2 + 1) \times 2$$

$$= 30$$

$$\text{Required probability} = \frac{30}{11 \times 11} = \frac{30}{121}$$

Q8

	A	B
E_1 : A is selected	3 W	3 W
	7R	2R

 E_2 : B is selected

E : white ball is drawn

$$P(E_1/E) =$$

$$\frac{P(E) \cdot P(E/E_1)}{P(E_1) \cdot P(E/E_1) + P(E_2) \cdot P(E/E_2)} = \frac{\frac{1}{2} \times \frac{3}{10}}{\frac{1}{2} \times \frac{3}{10} + \frac{1}{2} \times \frac{3}{5}}$$

$$= \frac{3}{3+6} = \frac{1}{3}$$

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Solutions

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Q9

Probability of drawing first red and then white

$$= \frac{10}{75} \times \frac{30}{75} = \frac{4}{75}$$

Q10

3 bad apples, 15 good apples.

Let X be no of bad apples

$$\text{Then } P(X=0) = \frac{{}^{15}C_2}{{}^{18}C_2} = \frac{105}{153}$$

$$P(X=1) = \frac{{}^3C_1 \times {}^{15}C_1}{{}^{18}C_2} = \frac{45}{153}$$

$$P(X=2) = \frac{{}^3C_2}{{}^{18}C_2} = \frac{3}{153}$$

$$E(X) = 0 \times \frac{105}{153} + 1 \times \frac{45}{153} + 2 \times \frac{3}{153} = \frac{51}{153}$$

$$= \frac{1}{3}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= 0 \times \frac{105}{153} + 1 \times \frac{45}{153} + 4 \times \frac{3}{153} - \left(\frac{1}{3}\right)^2$$

$$= \frac{57}{153} - \frac{1}{9} = \frac{40}{153}$$

Q11

Let probability of tail is $\frac{1}{3}$ $\Rightarrow$ Probability of getting head = $\frac{2}{3}$ $\therefore$ Probability of getting 2 tails and 1 head

$$= \left(\frac{1}{3} \times \frac{2}{3} \times \frac{1}{3}\right) \times 3$$

$$= \frac{2}{27} \times 3$$

$$= \frac{2}{9}$$

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